Feeder contact verification

FCE12 • Native numerical method and transfer-subcycle results

Configuration: unchanged FM06-R1 reference geometry. Native solver: Project Chrono 10.0.0, rigid nonsmooth contact, Bullet collision and PSOR. Internal units are kg, mm and s; published trajectories, loads and work are converted to SI.

Study question

Does the collision representation resolve the actual CAD contact geometry at the relevant scale, and which transfer results remain sensitive to numerical settings? The study separates analytical calibration, geometric resolution and dynamic convergence.

Check	Outcome
Planar contact calibration	Both sizes detect imposed 10 µm–1 mm overlaps; largest checked reported-gap error 4.28 nm. General curved-surface accuracy is not inferred.
Unit conversion	No-contact controller replay agrees within 1e-6 in checked SI outputs.
Native case matrix	Ten nominal/friction/gravity cases reach 26.5 s; injected jam and stuck-pin cases abort at 11 s.
Full native-solid pose audit	84 targeted poses; two residual intersections admit a nonintersecting pose after 1 µm Y translation.
Final-pose and force refinement	Provisional 50 µm / 0.05 deg final-pose screen is not met. Impulsive contact peaks are not design loads.

These are computational method results. The sequence begins with the storage gate open and ends after capture and fork withdrawal. Full launch/empty return/reset, installed actuation and physical correlation are separate requirements. No feeder architecture is selected by this study.

1. Geometry and unit scaling

The native payload is partitioned into nine convex solids with disjoint interiors and exact B-rep union equivalence. The rounded rail and recessed-panel geometry is retained. Each mechanism is partitioned into orthogonal native material cells: 64 boxes per payload size within the guarded contact domain. Full original body mass, centre of gravity and inertia are retained. Nonplanar retained mechanism material is rejected rather than approximated by a box.

The payload domain is X=[–12, L+12] mm and Z=[14, 128] mm; Y coverage is [–500, 500] mm. Initial receiver material below14 mm is retained down to8 mm to cover its12 mm upward stroke. Leaving the X/Z domain rejects the dynamics. The independent audit uses the complete original B-reps, including material outside the collision domain.

Quantity	Native unit	Conversion to SI
Position / velocity	mm / mm·s■¹	divide by 1000
Mass	kg	unchanged
Inertia	kg·mm²	divide by 10■
Force / impulse	mN / mN·s	divide by 1000
Mechanical work	µJ	divide by 10■
Angle / angular velocity	rad / rad·s■¹	unchanged

The assumed collision envelope is5 µm, safe margin zero. The same physical numerical settings are converted consistently: gravity, actuator limits, guide resistance, penetration recovery speed and inertia. The 1 ms controller period and dimensionless contact coefficients are unchanged. The no-contact unit-invariance check isolates conversion correctness from collision behavior.

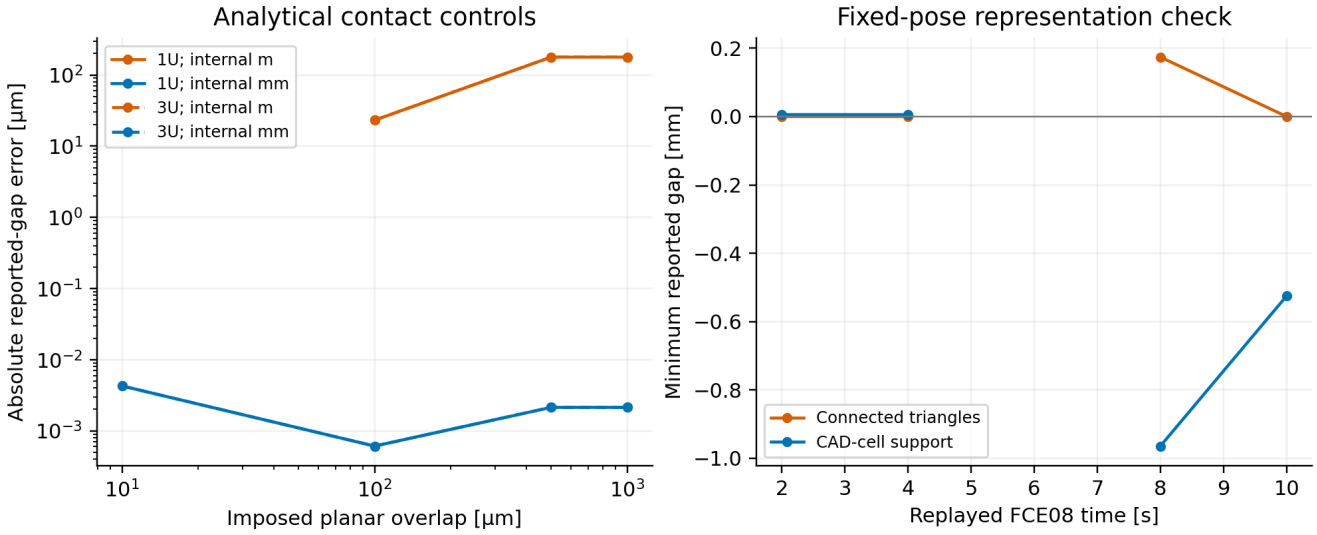


Figure 1. Analytical planar-gap controls and replayed fixed poses. The planar controls compare native coordinate scales. The replay uses actual captured FCE08 poses with the same material geometry; negative gap denotes detected penetration. Missing contacts are omitted from the plotted gap series.

2. Native case matrix

Case	Payload / condition	Step [µs]	Exit	Work [J]
01	1U / step refinement	500	0	1.91996
02	1U / step refinement	250	0	1.91996
03	1U / step refinement	125	0	1.91996
04	3U / step refinement	500	0	1.92209
05	3U / step refinement	250	0	1.92229
06	3U / step refinement	125	0	1.92254
07	3U / 4000 iterations	250	0	1.92238
08	3U / friction 0	500	0	1.92389
09	3U / friction 0.3	500	0	1.92242
10	3U / 6 kg, 1 g rigid	500	0	2.34258
11	3U / injected jam	500	11	0.23452
12	3U / injected stuck pin	500	11	0.96105

Exit 0 identifies the programmed endpoint; 11 identifies a commanded abort. Exit codes do not measure physical reliability. Mechanical work excludes motor/inverter losses, static holding power and the installed energy source. The 6 kg gravity case is rigid and does not include elastic tine/root/frame deflection.

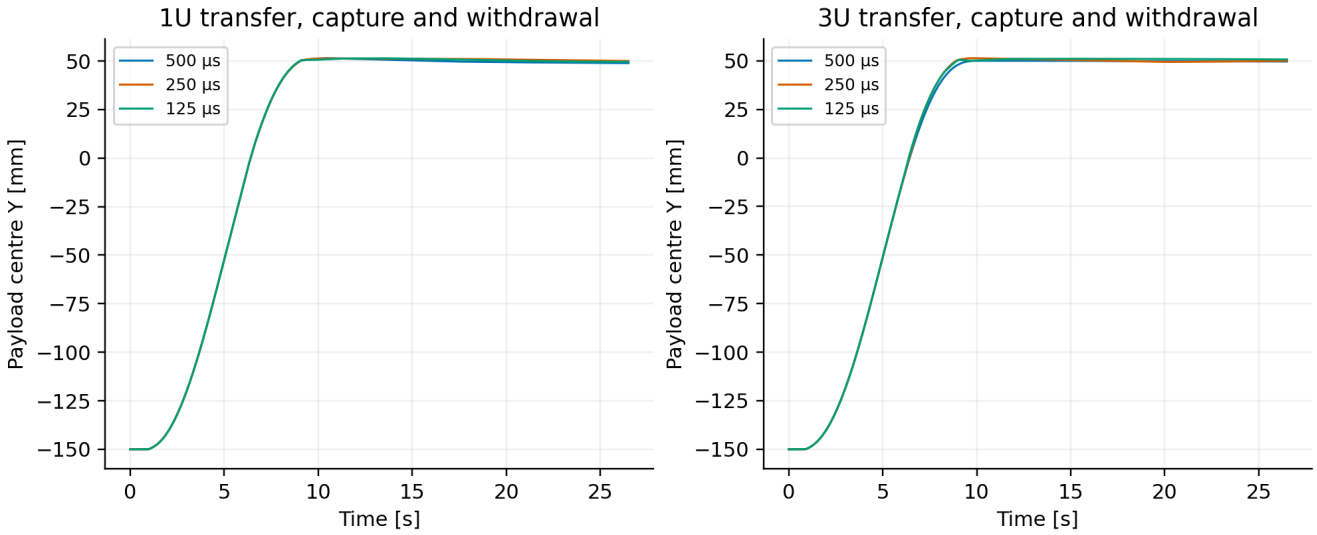


Figure 2. Native transfer/capture/withdrawal histories for 1U and 3U at three integration steps. Similar travel histories do not establish agreement in final six-degree-of-freedom state.

3. Numerical sensitivity and geometric resolution

Payload	250 → 125 µs COM change	Orientation change	Work relative change
1U	1.508584 mm	0.639678 deg	0.000276 %
3U	1.466380 mm	0.232303 deg	0.012691 %

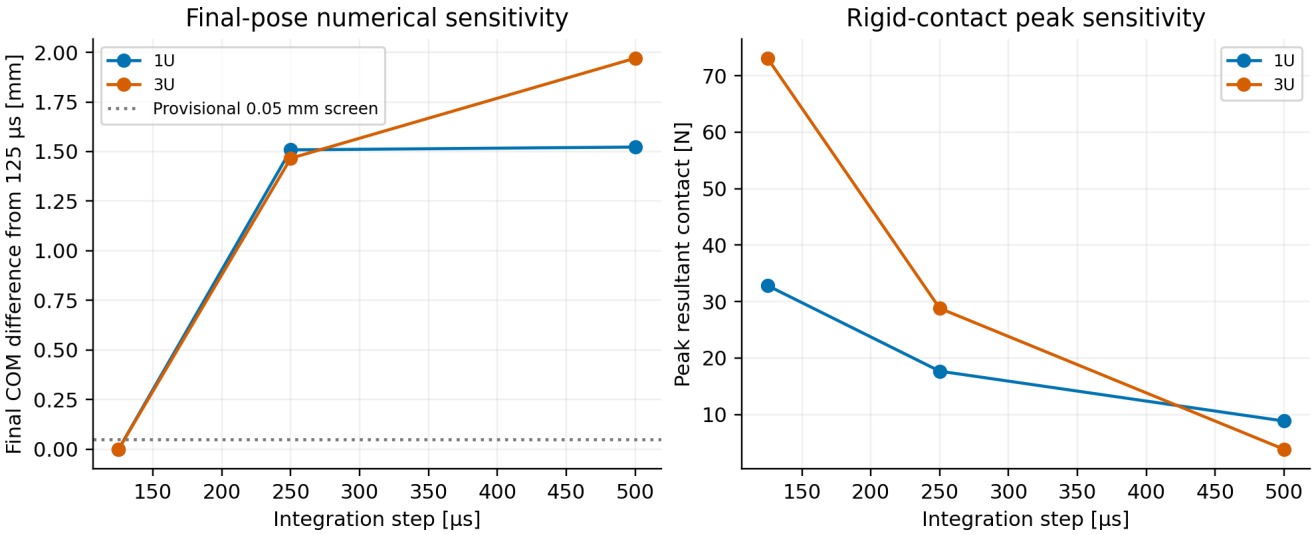


Figure 3. Final-pose and rigid-contact peak sensitivity. The 125 µs trace is a numerical comparison reference. The provisional 0.05 mm screen is not met; the peak contact values are not accepted structural loads.

Independent native-solid checks cover 12 targeted times in each of seven nominal/iteration cases, including the earlier transfer-interference times. Two 3U 500 µs poses retain positive material intersections: maximum 0.000124 mm³. Their original positions remain in NATIVE_POSE_AUDIT.json. A separate check translates the payload by 1 µm in Y and verifies zero positive intersections with all original components. This establishes a nearby nonintersecting pose for those residuals; it does not modify the trajectories or prove continuous nonpenetration.

The planar calibration and sampled full-solid checks support this collision representation at the declared numerical scale. Final-state sensitivity remains relevant because the modeled payload is loosely retained rather than correlated against a measured preloaded contact system. Precision, force ratings and wear/reliability require further contact/joint evidence.

4. Traceability and engineering boundary

Frozen campaign records retain exact command arguments, source/executable identities, solver outputs, CSV histories, metrics and hashes. The exact executed source is preserved under captured/. Formatted root sources are verified to have equivalent non-comment tokens. CMake builds the readable sources; a replay writes a separate output directory. MANIFEST.json identifies every distributed file.

Independent checks

• Native B-rep volume equivalence and disjoint payload-cell interiors.

• Exact orthogonal mechanism-cell volume identities and guarded-domain definition.

• Fourteen analytical planar-gap controls; no-contact SI unit invariance.

• Twelve native dynamic cases and seven full native-solid pose reconstructions.

• 84 selected poses and explicit residual-resolution checks.

• Separate timestep, iteration, friction, gravity and injected-fault outcomes.

Engineering interpretation

This work closes a numerical representation question for the stated subcycle. It does not close source-rated actuation, joint flexibility, actual friction/compliance, payload interface approval, a complete release/reset mechanism, multi-cell installed integration or a seven-architecture downselect. Receiver and tine finite-element results belong to their separately stated fixtures. No physical prototype, measured precision, qualification or flight acceptance is represented.

All source files, local replay commands, scientific figures and machine-readable criteria are included with the report. Historical FCE08 and intermediate diagnostic results remain separately traceable; they are not relabeled as accepted current performance.